However, cost is a factor for long-distance communication.

A data centre is the data processing, routing and storage centre in a network. With data-intense applications and platforms coming online, increasing efficiency is becoming a more important aspect. With this, rack space, installation, cable pathways, cooling, maintenance and risk avoidance have become increasingly important considerations while planning and designing a data centre.

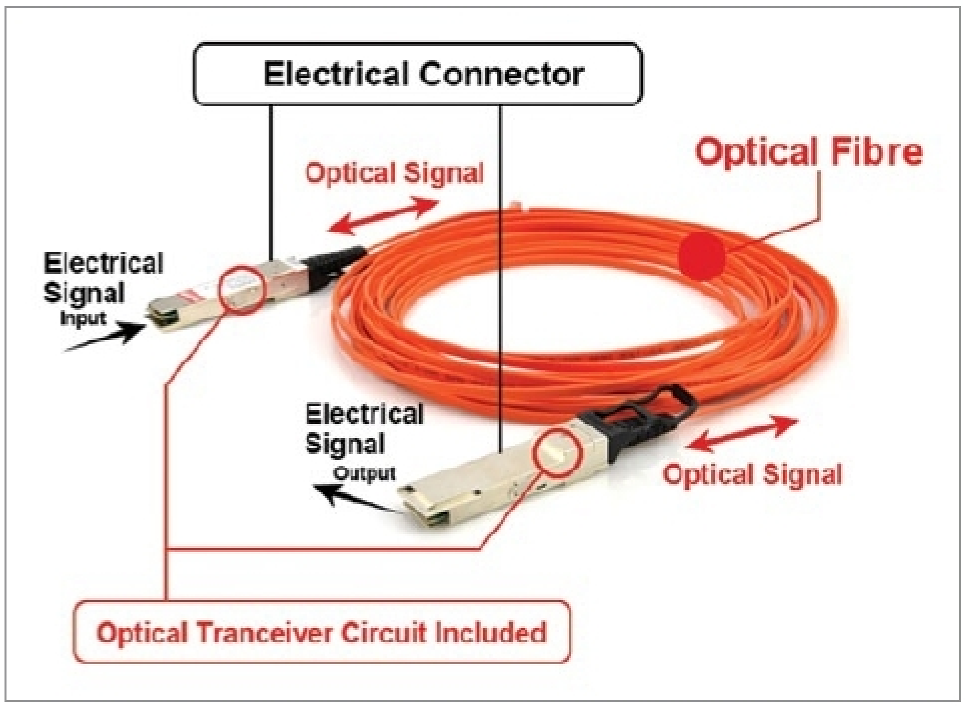

Fig. 1: Active optical cable structure (Credit: www.cables-solutions.com)

Data centre cabling infrastructure must be designed to move data as fast as possible with the lowest latency. This needs the fastest possible data communications links. As a result, next-generation applications such as cloud computing and virtualisation must be ensured early in the data centre cabling infrastructure design and planning process. Big data centre users want data rates of 10Gbps, 40Gbps and 100Gbps as fast as possible, and are pushing network standards committees and manufacturers to produce compatible products.

TABLE I
FIBRE TYPES AND THEIR TYPICAL REACH

Fibre types	Typical reach
Active optical cables (AOC)	• ≤100m
Duplex multimode fibre	• ≤100m to 180m • ≤220m to 500m
Parallel multimode fibre	• ≤100m • ≤400m
Duplex single mode fibre	• ≤500m to 2km • ≤10km • ≤20km to 40km
Parallel single mode fibre	• ≤500m to 2km

(Credit: www.finisar.com)

Role of optical-fibre for data centres

Establishing high-speed network connections is now a function of the availability of physical media like fibre rather than being dependent on more expensive proprietary network equipment. It is now almost as easy to set up a 100Gbps or faster optical connection across hundreds to a thousand kilometres or more as it is for a short few hundred metres between racks at an interconnect point.

With availability of fibre, data centres are no longer restricted to one or two centralised locations. Instead, these can be at any place that has a fibre connection, with information collected, stored and transferred to where it is needed for business processes or regulatory requirements.

The role of cable manufacturers is to adapt to new customer requirements in terms of density, bandwidth, latency and form factors. Cables with larger fibre counts, with small primary buffer diameters, will find more applications. Use of stranded ribbon cables allows operators faster installations. Cables with more accurate fibre length not only save money but also facilitate installation. Need for equalised optical length cables is already present for some fintech customers, whether that is standards lead or a government or financial authority requirement.

A fibre-optic cable delivers the fastest and cleanest connections. It is more reliable and future-proof than copper. It can save money in the long term, and only a basic understanding is required for a successful deployment.

Multi-mode fibre versus single-mode fibre

Multi-mode (MM) fibre can work over a range of 100m depending on the type of fibre chosen, but it is limited on power budget, which means the cable plant must be a low loss. It is commonly used for inter-office deployments and performs better in short distances. Due to its larger inner core, it can be lit by cheaper, less complex equipment than a single-mode (SM) fibre. It has been widely used for up to 10G, but beyond 10G, cost and cabling bulk become a challenge. So, it utilises parallel optics with 10G channels to build higher speeds.

MM runs out of bandwidth at 10G, so parallel optics with separate 10G channels are used at 40G and 100G. Four 10G channels for 40G and ten 10G channels for 100G are used to support equipment. MM links at 10G are also power-limited due to the limited modal bandwidth of MM fibre.

SM fibre is most commonly used by Internet service providers (ISPs) and telephone companies because of its ability to span long distances. The equipment needed to work with

TABLE II
DATA CENTRE FIBRE CABLING OPTIONS

Media	1Gbps	10Gbps	40Gbps	100Gbps
MM fibre	OM2/3/4	OM3/4	• OM3/4 • Parallel, eight fibres • WDM, two fibres, proposed	• OM3/4 • Parallel, 20 fibres • OM5 CWDM, two fibres
SM fibre	OS1/2	OS1/2	OS1/2, WDM	OS1/2, WDM
AOC (short)	NA	SFP+	QSFP+/SFP+	CXP

(Credit: www.thefoa.org)